



ADVANCEMENT IN ATTENDANCE TAKING PROCEDURE FOR STREAMLINED OPERATIONS

Authors

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INTRODUCTION

The primary goal of this project is to create an automated student attendance system based on facial recognition. The test pictures and training images of this suggested technique are restricted to frontal and upright facial images that only contain a single face to improve performance. To ensure no quality variation, the test photographs and training images must be taken using the same equipment. To be identified, the pupils must also register in the database. The user-friendly interface allows for immediate enrollment.

OBJECTIVE

The goal of this project is to create an automated student attendance system based on facial recognition. The following are expected results to meet the objectives: To identify the facial segment in the video frame.

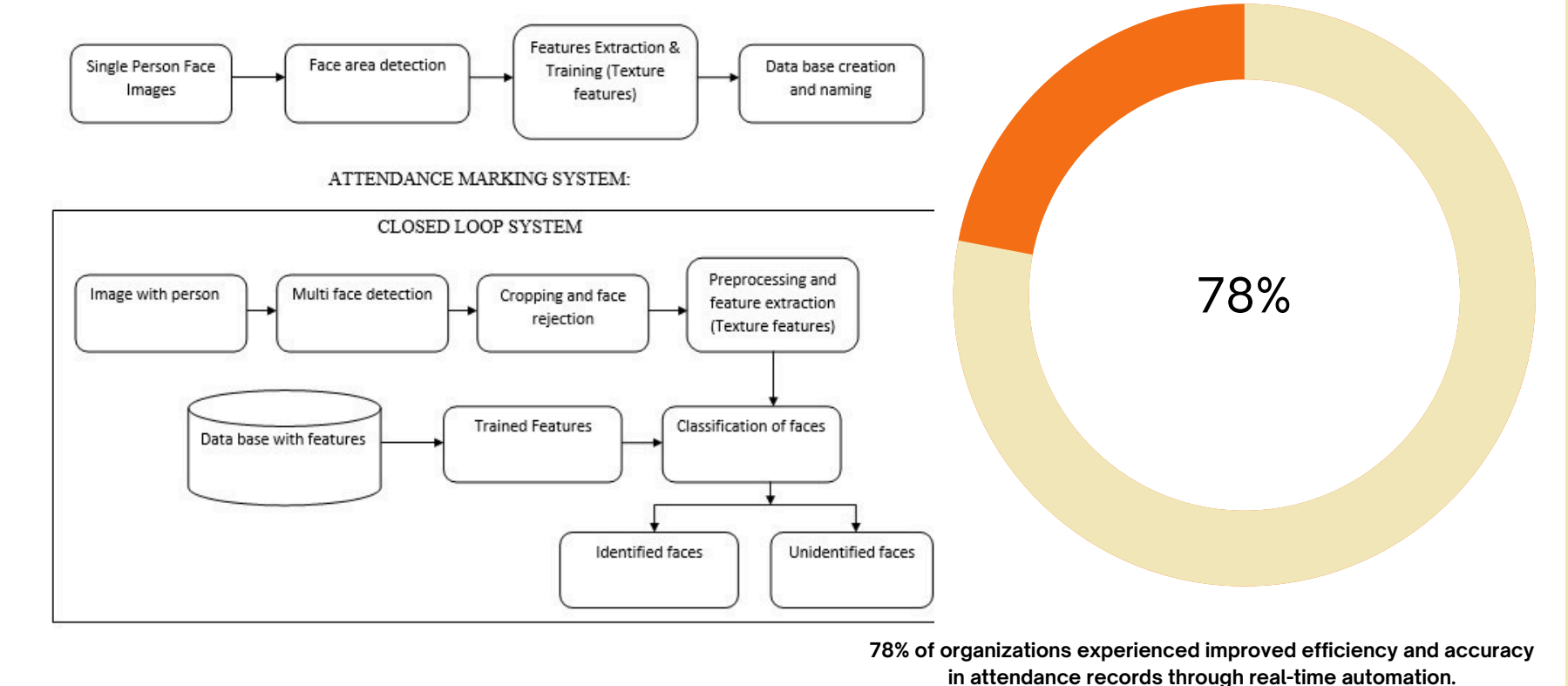
- To identify the useful aspects on the observed face.
- To categorize the traits in order to identify the observed face.
- To note the indicated student's attendance

METHODOLOGY&ARCHITECTURE

The method uses a facial recognition-based attendance system for students. The technique starts with the picture capturing utilizing an easy-to-use interface, then moves on to pre-processing the facial photos that were collected, feature extraction from those images, subjective selection, and finally categorization of the facial images to be recognized. This suggested technique computes both LBP and PCA feature extraction algorithms, then compares their performance. With this method, LBP is improved to lessen the lighting impact. Moreover, a subjective selection method that combines improved LBP and PCA is created to improve accuracy. The next sections will go through the specifics of each level.

FUTURE SCOPE

The future scope of this project includes numerous enhancements and applications. Scalability and integration with existing school management systems will enable seamless operation and accommodate larger databases. Real-time processing improvements will ensure quicker and more efficient recognition, beneficial for larger classrooms and events. Advanced security measures will be implemented to safeguard student data and prevent unauthorized access. Mobile applications and cloud-based solutions will provide remote access and management, enhancing flexibility. Incorporating advanced algorithms and deep learning techniques will increase accuracy and reliability, particularly in diverse environments. The system could also benefit from multimodal recognition, combining facial recognition with other biometric modalities like voice or fingerprint recognition to enhance verification processes. User experience can be further refined by improving the GUI and adding features such as automated reporting and notifications. Additionally, ongoing research on optimal parameters and techniques, such as the choice of radius in Local Binary Pattern (LBP), will continuously improve the system's effectiveness and efficiency.



REFERENCES

Prominent studies and resources have significantly contributed to the development and understanding of the face recognition student attendance system. Robinson-Riegler and Robinson-Riegler (2008) offer insights into cognitive psychology, while Margaret Rouse (2012) provides foundational definitions of facial recognition. Further advancements are detailed by Solon (2017) and Silk (2017), with Zhao et al. (2003) and Ojala et al. (2002) discussing critical facial recognition algorithms. Studies on automated attendance systems, such as those by Wagh et al. (2015) and Patil and Bangar (2017), highlight practical applications of these technologies. These references collectively underpin the system's technological framework and practical implementation.

RESULTS

The recommended technique employs a MATLAB GUI to create a user-friendly face recognition student attendance system. The interface includes several buttons, each serving a distinct function: the register button facilitates student enrollment, the start button activates the camera and automatically performs facial recognition, and the update button trains the latest photos added to the database. Additionally, the browse and recognise buttons allow users to navigate through face images in a selected database, identify a chosen image, and assess system performance. The system utilizes an enhanced Local Binary Pattern (LBP) method with a radius of two, with a discussion provided on the choice of radius size.

